# Topic 2: Making Your First API Calls

N8N Hands-On API Integration Course • Step-by-Step Practical Guide

> **Overview:** This step-by-step guide walks you through executing GET, POST, PUT, PATCH, and DELETE requests using the ReqRes public mock API sandbox ( `https://reqres.in` ). You will configure query parameters, customize request headers, format JSON request bodies, and parse successful status codes (200, 201, 204) in both Postman and n8n.

## Step 1: Understand the ReqRes Sandbox Endpoints

For this topic, we will use ReqRes ( `https://reqres.in` ), which is a hosted mock server that supports all standard CRUD API requests:

- **Base URL:** `https://reqres.in`
- **Inbound Methods Supported:** `GET` , `POST` , `PUT` , `PATCH` , `DELETE`
- **Response Format:** `application/json`

## Step 2: Configure Query Parameters (GET Request)

Query strings append search variables to your URL. We will fetch page 2 of the mock users list:

- **Target Endpoint:** `GET https://reqres.in/api/users?page=2`

### In Postman:

1. Open a new tab, select `GET` method, and enter the URL: `https://reqres.in/api/users` .
2. Select the **Params** tab below the URL bar.
3. In the key-value table, enter:
   - **Key:** `page`
   - **Value:** `2`
4. Notice the URL automatically changes to `https://reqres.in/api/users?page=2` .
5. Click **Send** and check that the status code is `200 OK` .

### In n8n:

1. Open your n8n canvas. Add a **Manual Trigger** node.
2. Connect it to a new **HTTP Request** node named `HTTP_GetUsersPage2` .
3. Set the parameters:
   - **Method:** `GET`
   - **URL:** `https://reqres.in/api/users`
   - **Send Query Parameters:** Toggle to `Yes` .
   - **Parameters:** `Name: page | Value: 2`
4. Click **Test step** and verify the JSON list of page 2 users returns in the right panel.

## Step 3: Create a Record (POST Request with JSON Body)

POST requests send data bodies to the server to create a new resource:

- **Target Endpoint:** `POST https://reqres.in/api/users`
- **JSON Payload:**

```
{
  "name": "Alice Smith",
  "job": "Software Engineer"
}
```

### In Postman:

1. Open a new tab, select `POST`, and enter `https://reqres.in/api/users`.
2. Under the URL bar, select the **Body** tab.
3. Select the **raw** radio button, and select **JSON** from the dropdown menu on the right.
4. Paste the JSON payload into the editor and click **Send**.
5. Verify the status code is `201 Created` with generated `id` and `createdAt` timestamp.

### In n8n:

1. Connect a new **HTTP Request** node named `HTTP_CreateUser`.
2. Set parameters: Method: `POST`, URL: `https://reqres.in/api/users`, Send Body: `Yes`, Content Type: `JSON`, Specify Body: `Using JSON`.
3. Paste the JSON payload, click **Test step**, and verify status code 201.

## Step 4: Update a Record (PUT Request)

PUT is used to replace an entire existing resource, whereas PATCH is used for partial updates:

- **Target Endpoint:** `PUT https://reqres.in/api/users/2`
- **JSON Payload:**

```
{
  "name": "Alice Jones",
  "job": "Lead Engineer"
}
```

### In n8n:

1. Add a new **HTTP Request** node named `HTTP_UpdateUser`.
2. Set parameters: Method: `PUT`, URL: `https://reqres.in/api/users/2`, Send Body: `Yes`, Specify Body: `Using JSON`.
3. Paste the JSON payload and click **Test step**. Confirm status code `200 OK` and `updatedAt` timestamp.

## Step 5: Delete a Record (DELETE Request)

DELETE removes a resource from the server:

- **Target Endpoint:** `DELETE https://reqres.in/api/users/2`

### In n8n:

1. Add a new **HTTP Request** node named `HTTP_DeleteUser`.
2. Set parameters: Method: `DELETE`, URL: `https://reqres.in/api/users/2`.
3. Click **Test step**. Verify status code `204 No Content` (or `200 OK`).

## Step 6: Build and Execute the Combined CRUD Workflow

1. Arrange your n8n nodes in a single line on the canvas:

```
Manual_Trigger -> HTTP_GetUsersPage2 -> HTTP_CreateUser -> HTTP_UpdateUser -> HTTP_DeleteUser -> Code_AuditSummary
```

2. Rename the workflow to: **N8N CRUD Verification Workflow**.
3. Click **Save** in the top-right corner.
4. Click **Test workflow** and confirm that all nodes execute successfully displaying green checkmarks.

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